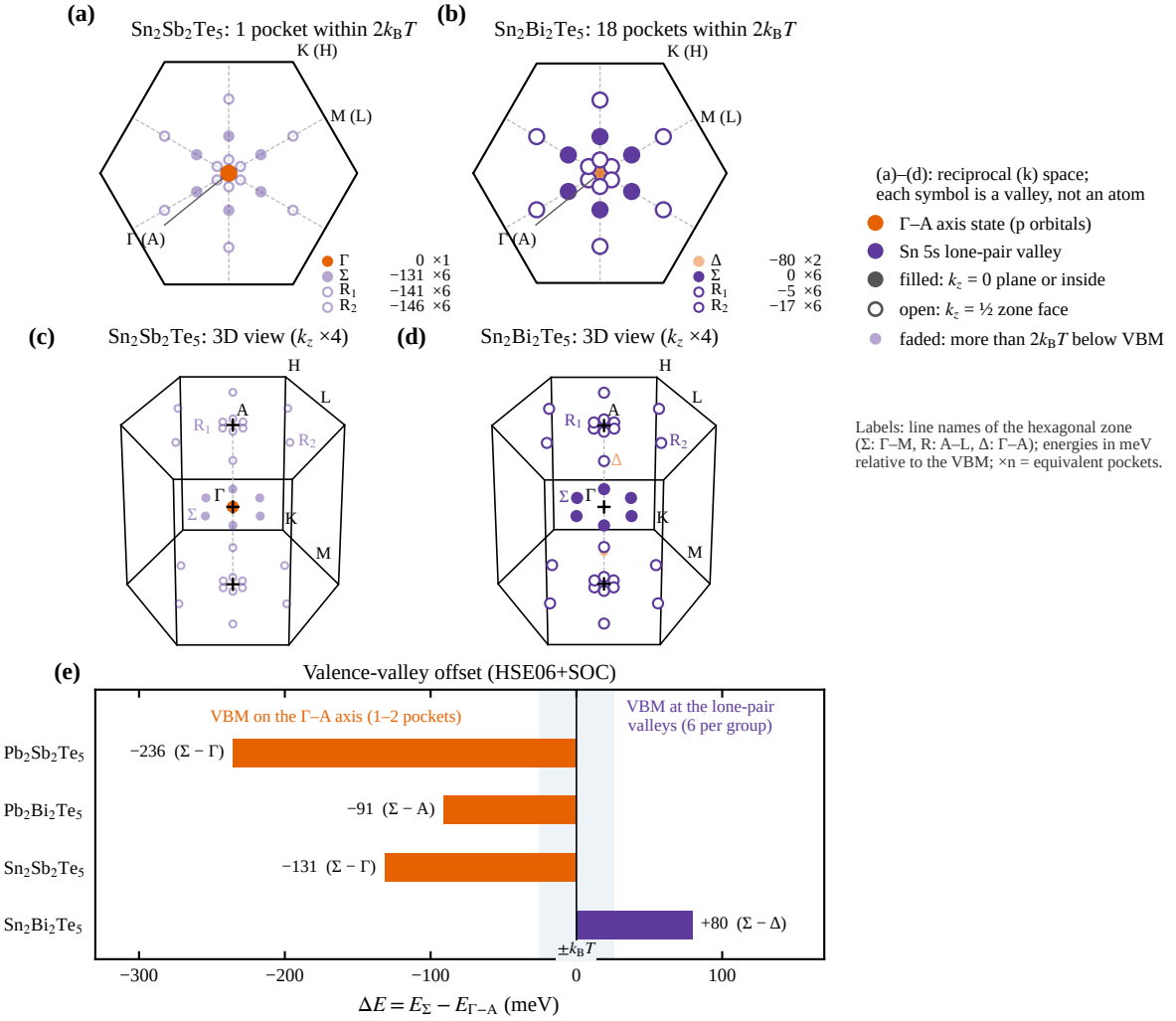
(a)
Sn2Sb2Te5: 1 pocket within 2kBT
K (H)
M (L)
Γ (A)
Γ 0 ×1
Σ −131 ×6
R1 −141 ×6
R2 −146 ×6
(b)
Sn2Bi2Te5: 18 pockets within 2kBT
K (H)
M (L)
Γ (A)
Δ −80 ×2
Σ 0 ×6
R1 −5 ×6
R2 −17 ×6
(a)–(d): reciprocal (k) space;
each symbol is a valley, not an atom
Γ–A axis state (p orbitals)
Sn 5s lone-pair valley
filled: kz = 0 plane or inside
open: kz = ½ zone face
faded: more than 2kBT below VBM
Labels: line names of the hexagonal zone
(Σ: Γ–M, R: A–L, Δ: Γ–A); energies in meV
relative to the VBM; ×n = equivalent pockets.
(c)
Sn2Sb2Te5: 3D view (kz ×4)
H
L
A
R1
R2
Γ
Σ
K
M
(d)
Sn2Bi2Te5: 3D view (kz ×4)
H
L
A
R1
R2
Δ
Σ
Γ
K
M
(e)
Valence-valley offset (HSE06+SOC)
VBM on the Γ–A axis (1–2 pockets)
VBM at the lone-pair
valleys (6 per group)
Pb2Sb2Te5
−236 (Σ − Γ)
Pb2Bi2Te5
−91 (Σ − A)
Sn2Sb2Te5
−131 (Σ − Γ)
Sn2Bi2Te5
+80 (Σ − Δ)
±kBT
−300
−200
−100
0
100
ΔE = EΣ − EΓ−A (meV)